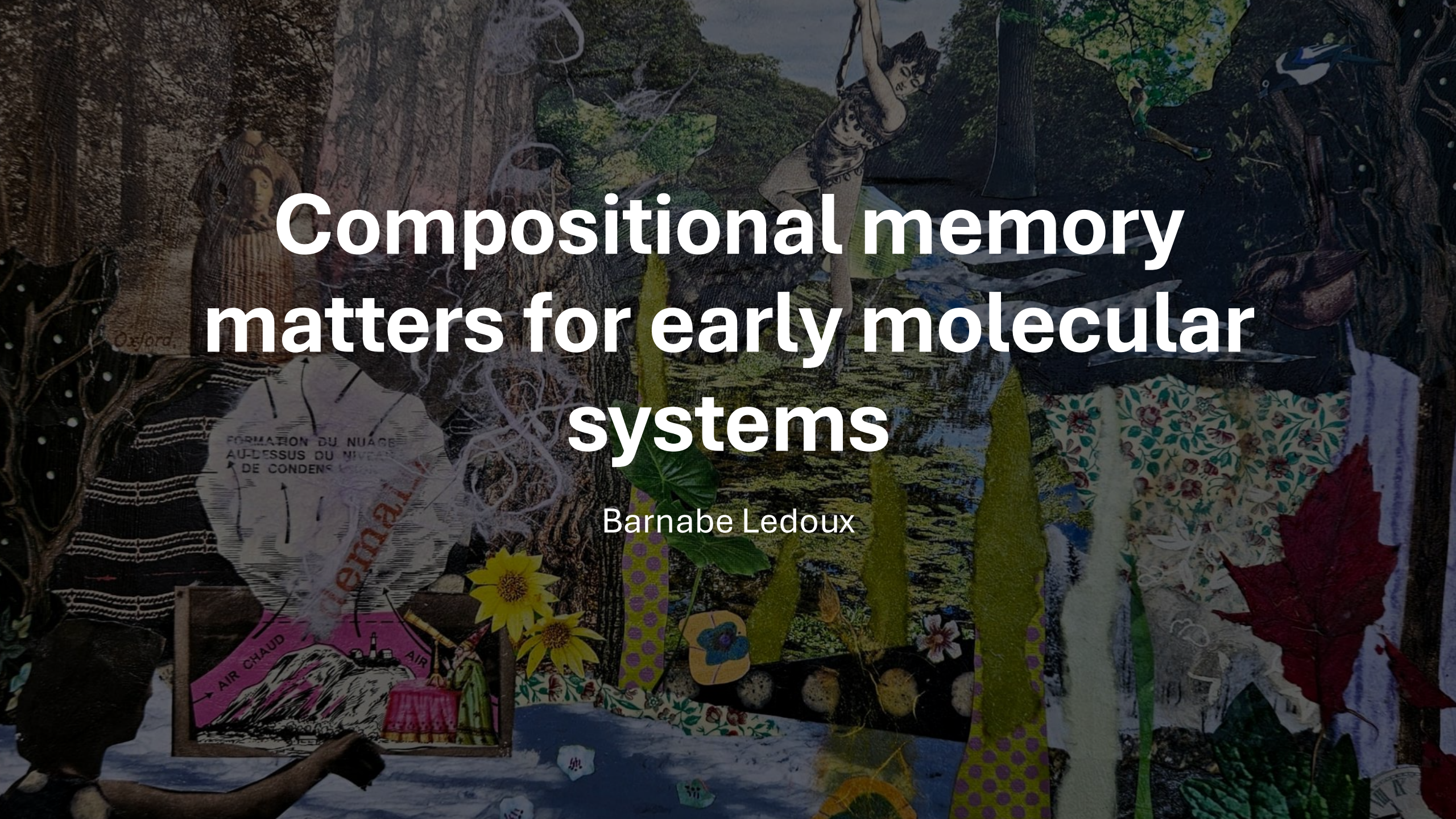
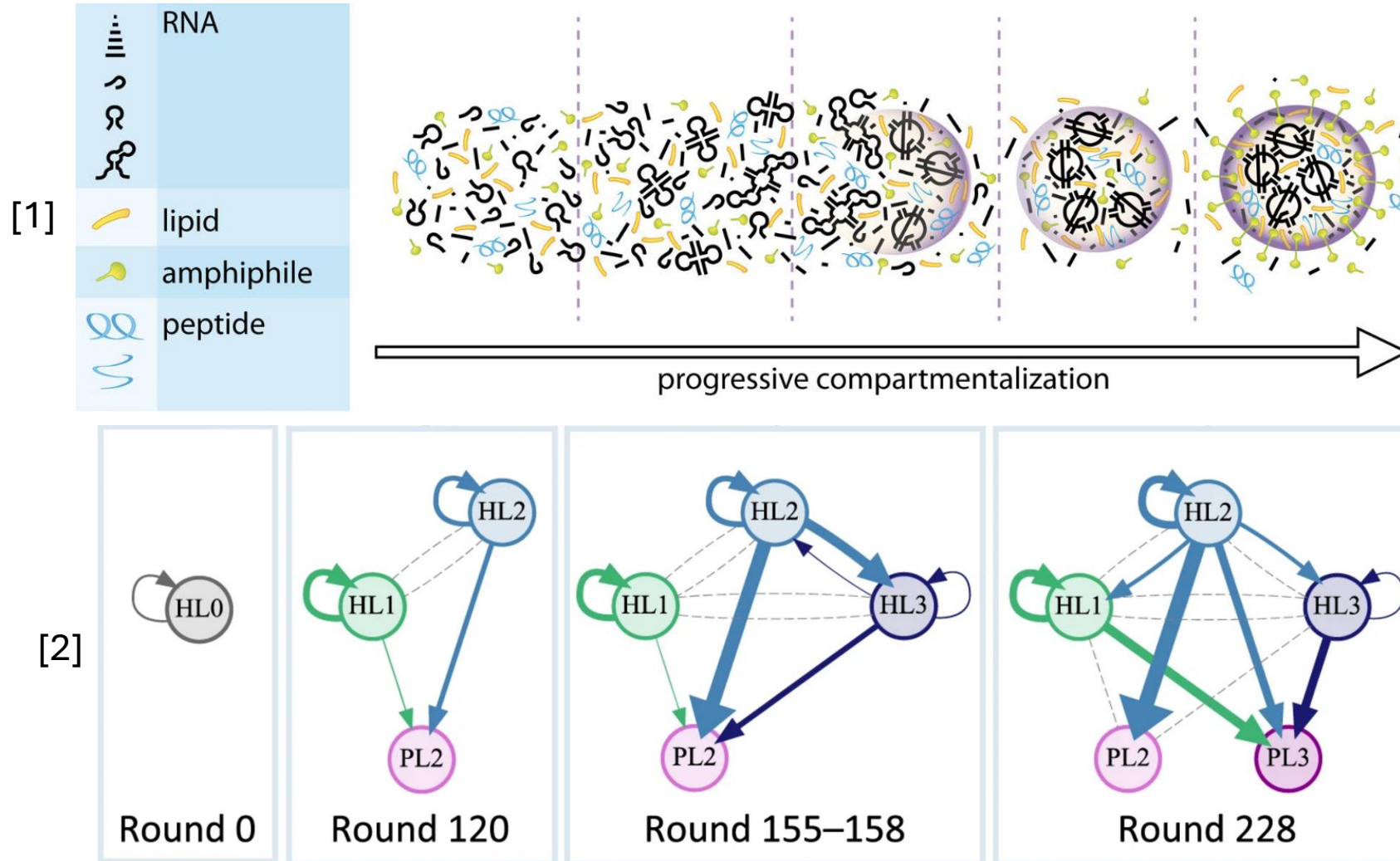




Compositional memory matters for early molecular systems

Barnabe Ledoux

Evolution of complexity



[1] P.Nghe, *A stepwise emergence of evolution in the RNA world*, FEBS Letters, 10.1002/1873-3468.70065 (2025)

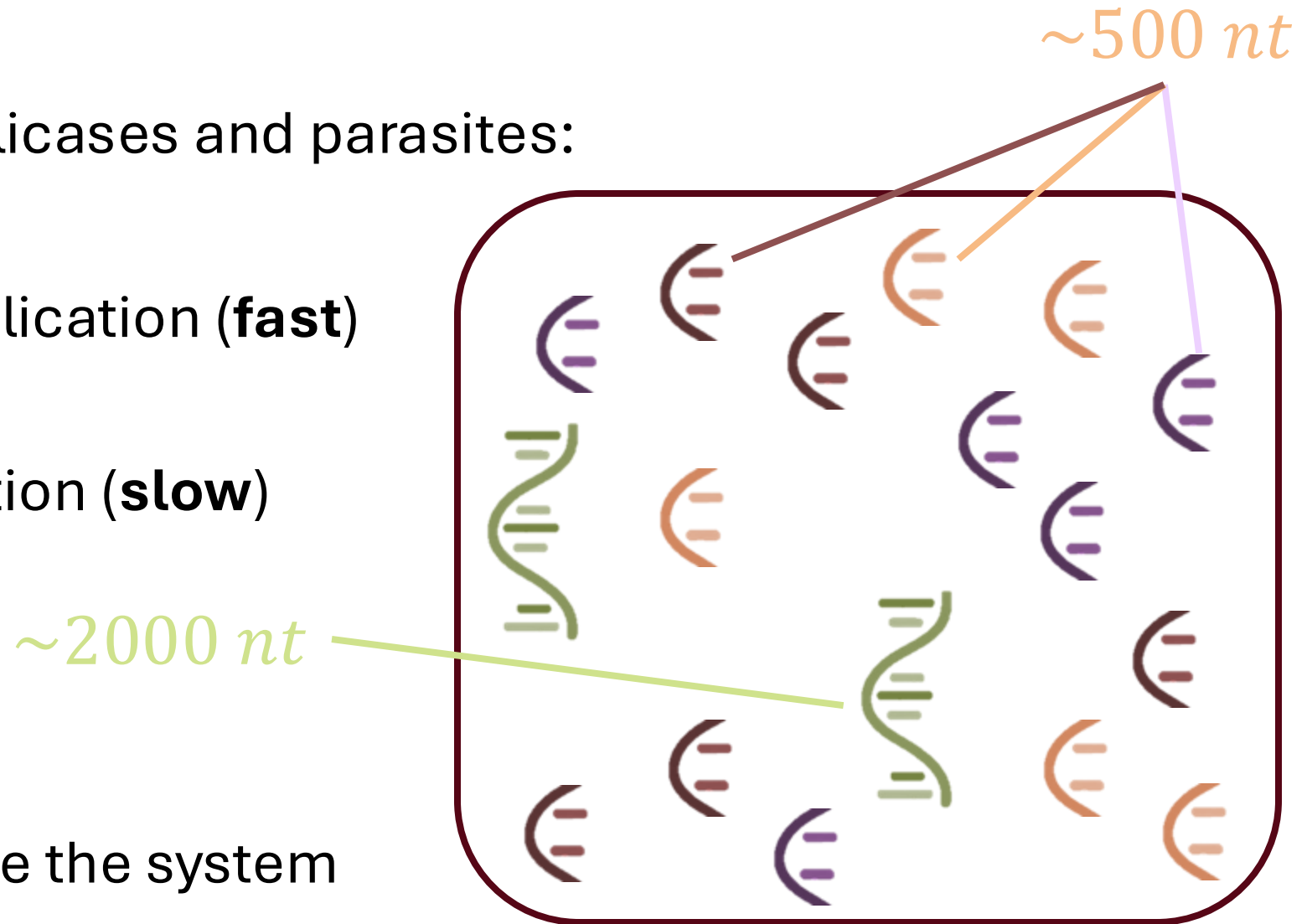
[2] R Mizuuchi, T Furubayashi, N Ichihashi, *Evolutionary transition from a single RNA replicator to a multiple replicator network*. Nat. Commun. 13, 1460 (2022)

The need for compartments

In a **pool** with replicases and parasites:

- Parasite replication (**fast**)
- Self replication (**slow**)

⇒ Parasites invade the system

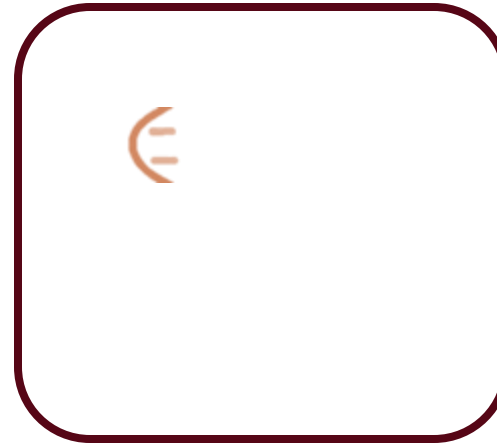


The need for compartments

With compartments:

- Parasite replication (**fast**)
- Self replication (**slow**)
- Parasites alone cannot replicate

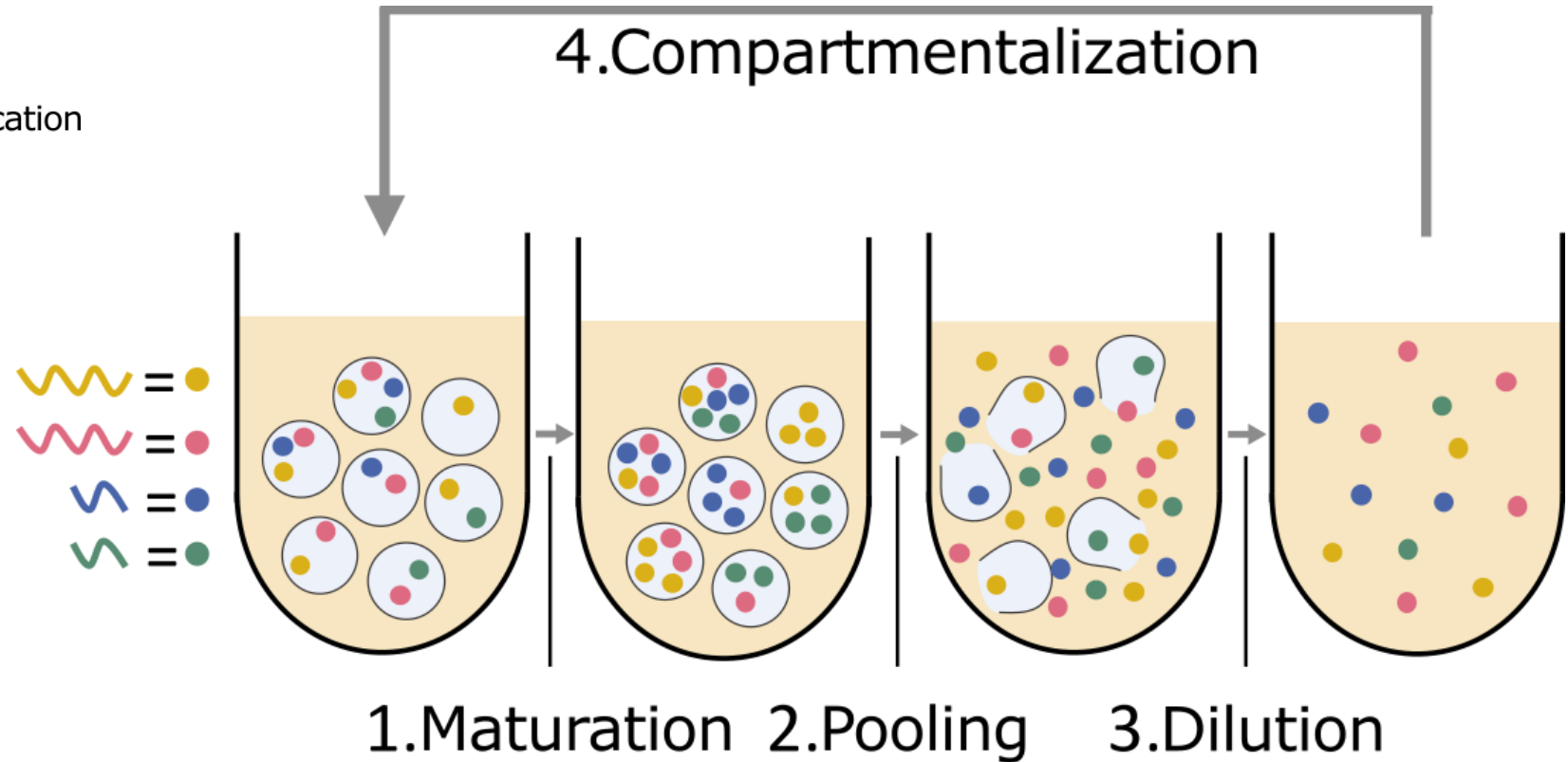
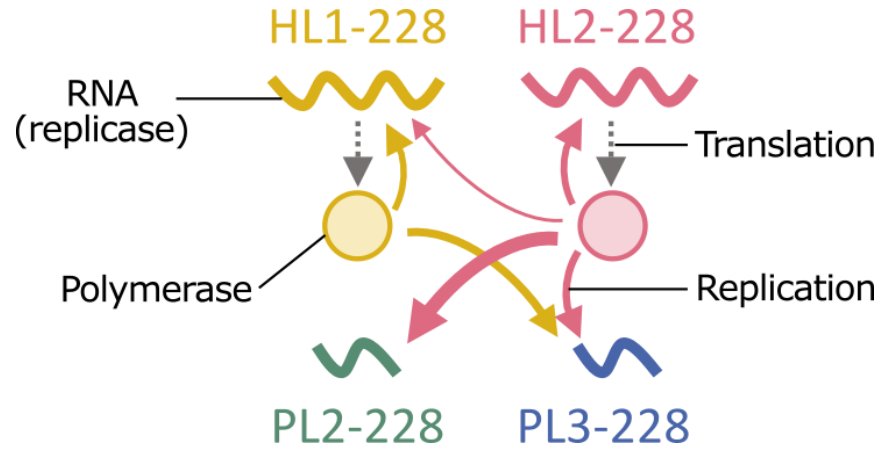
Isolated parasite



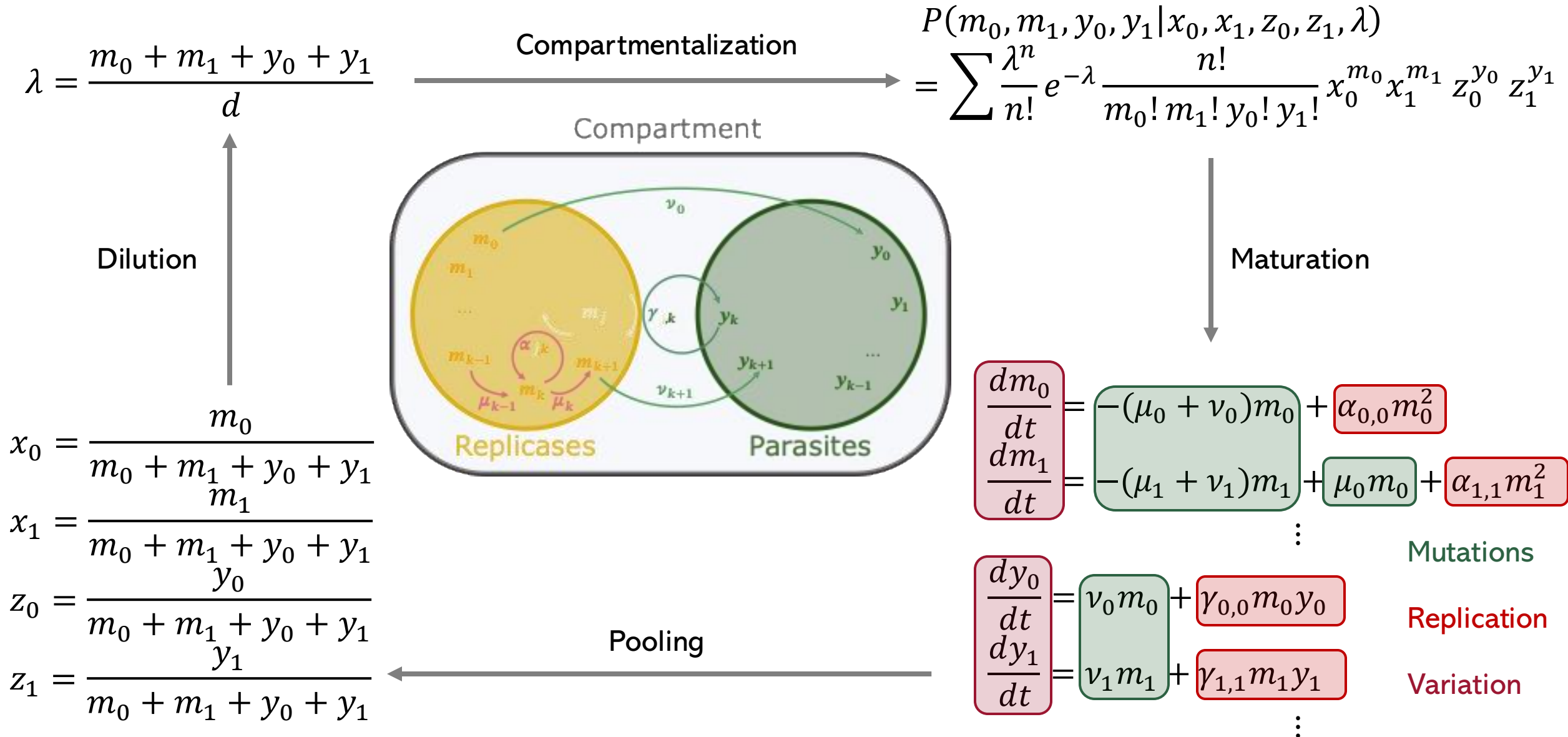
Modeling dynamics



Transient compartmentalization with natural selection and mutations



Evolution equations

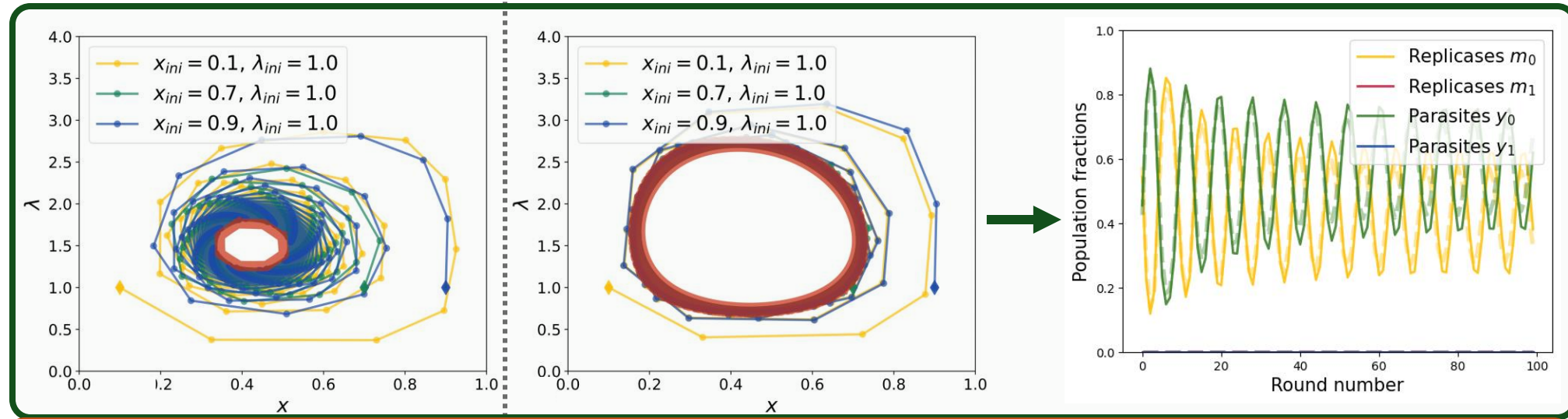


Oscillatory dynamics

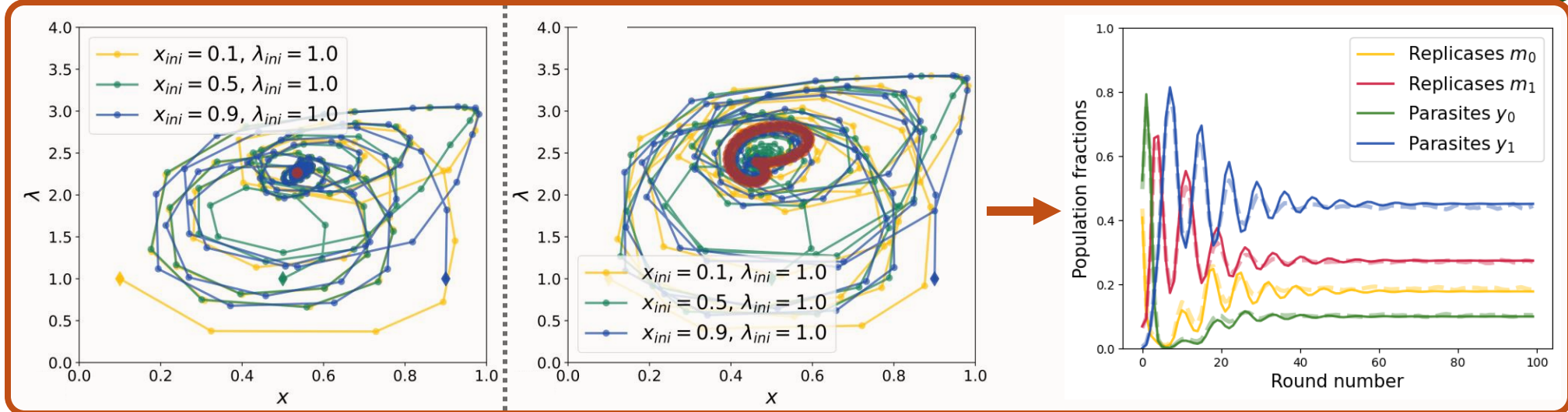
$$\frac{K}{\bar{d}} = 2.9 \quad \text{Carrying capacity} \quad \frac{K}{\bar{d}} = 3.2$$

Dilution

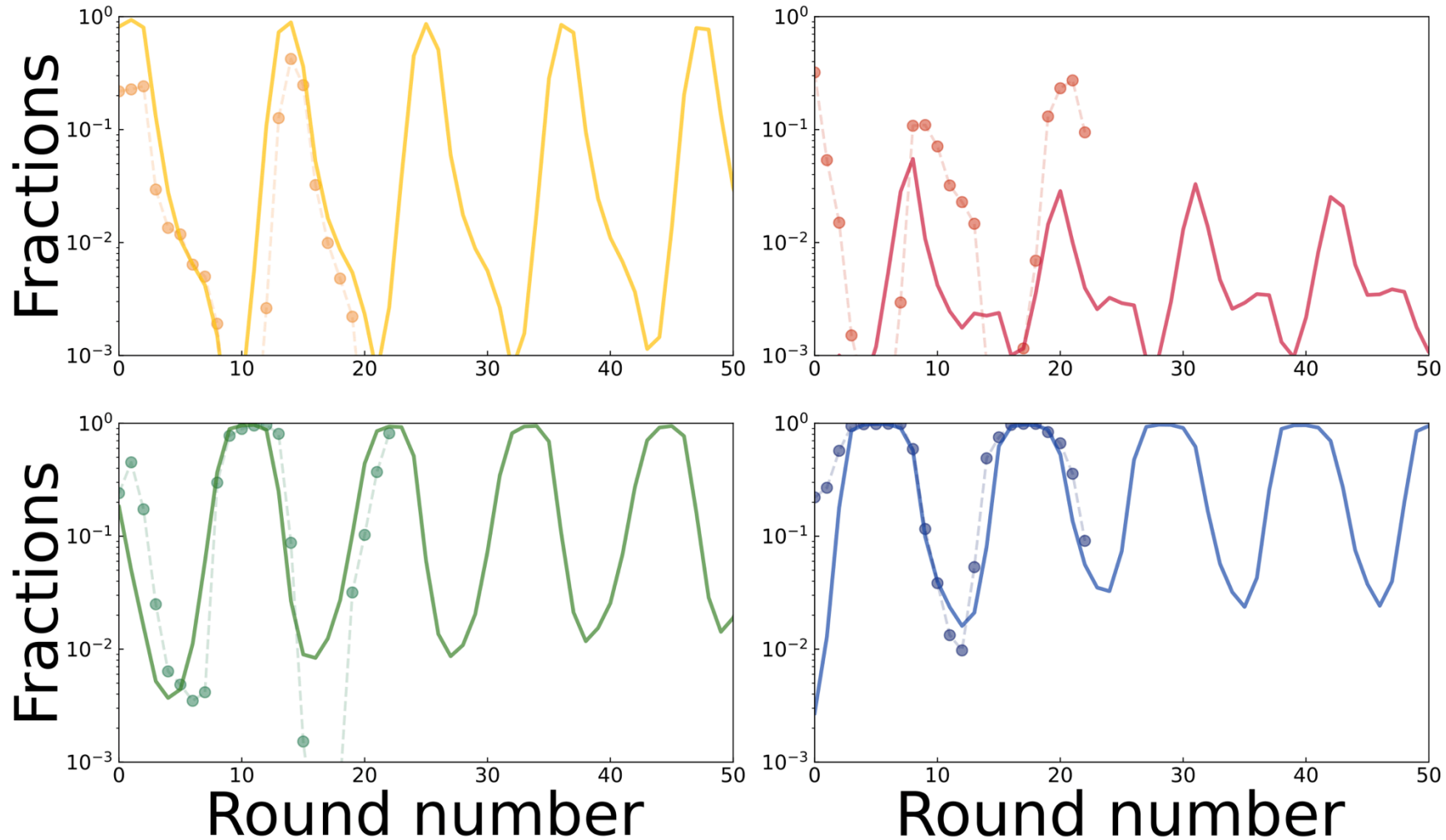
1 parasite
1 replicase



2 parasites
2 replicases



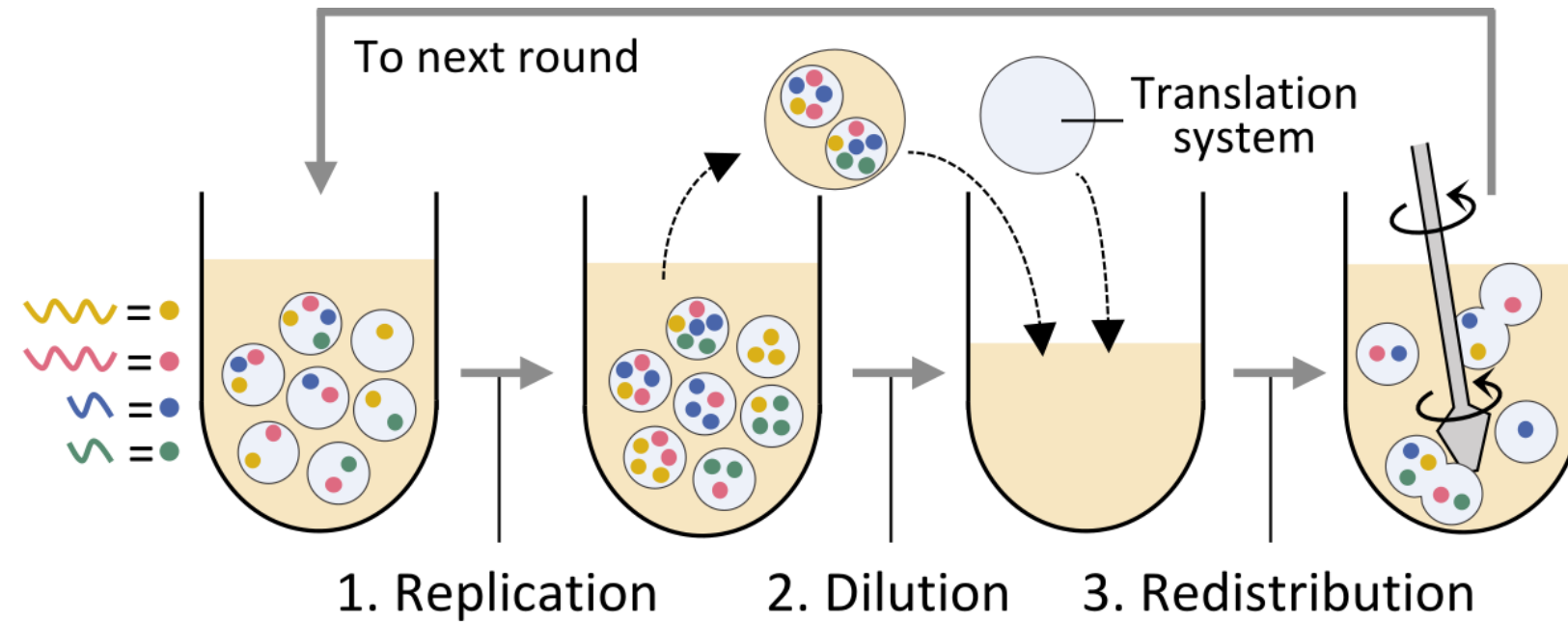
Comparison to experiments



Stirring



Stirring in serial transfer



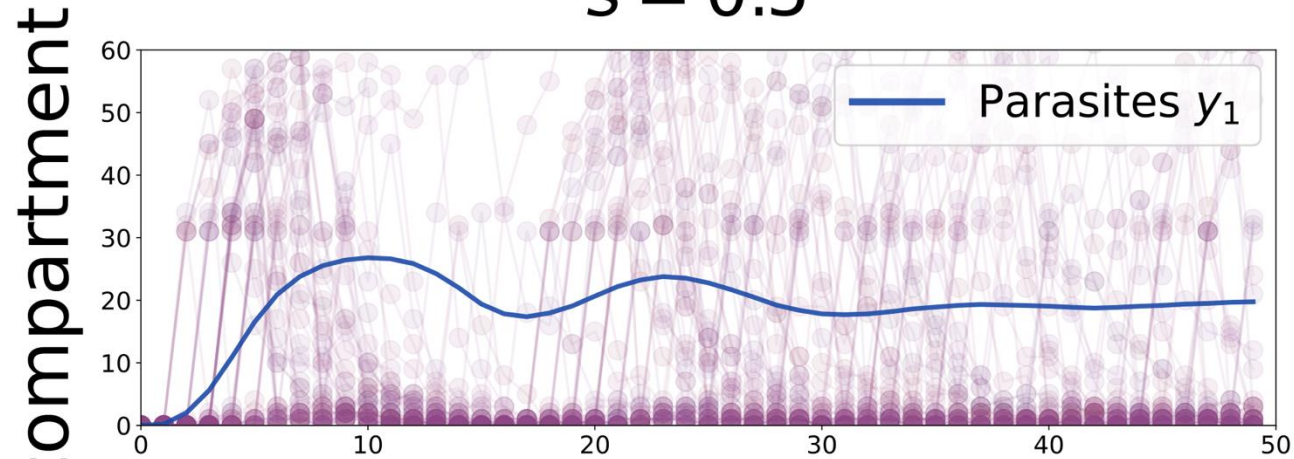
Memory of the compartment

$$x_0^{(i)} = \frac{(1-s)m_0^{(i)}}{(1-s)n^{(i)}} + \frac{\frac{s}{n_{cell}-1} \sum_j m_0^{(j)}}{\frac{s}{n_{cell}-1} \sum_j n^{(j)}}$$

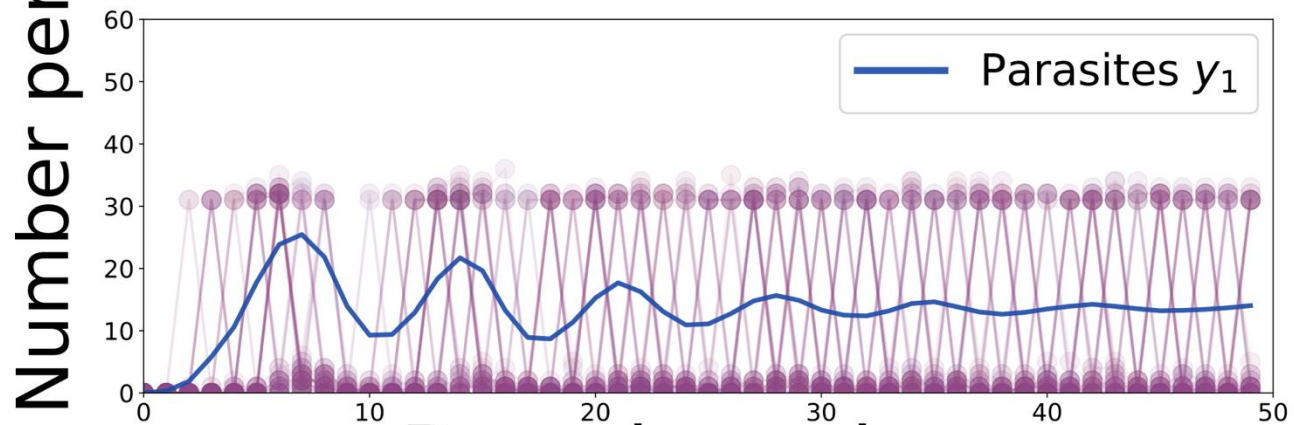
Mean field effect

Single compartment evolution

$s = 0.5$



$s = 0.99$



Increasing stirring

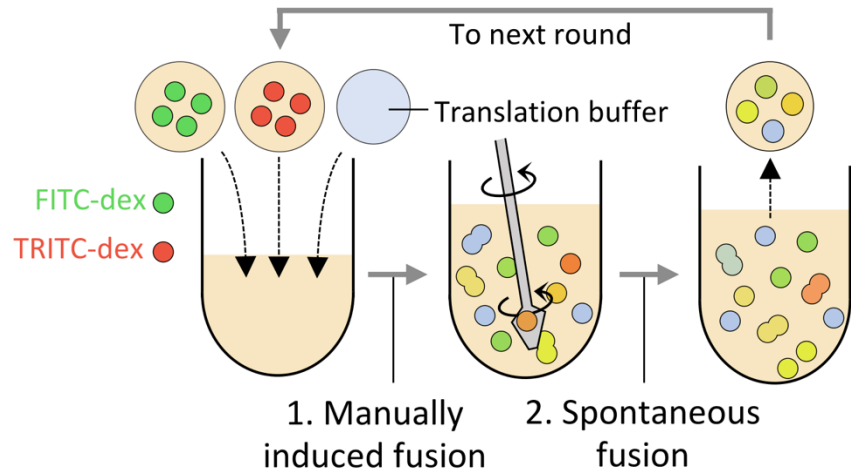


Increasing heterogeneity

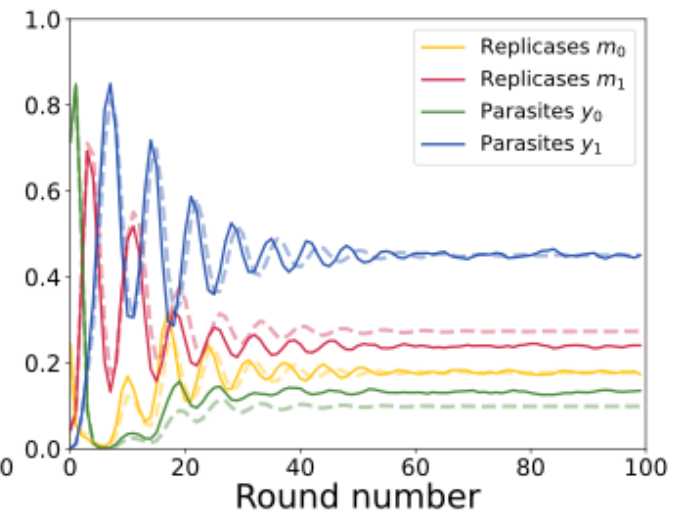
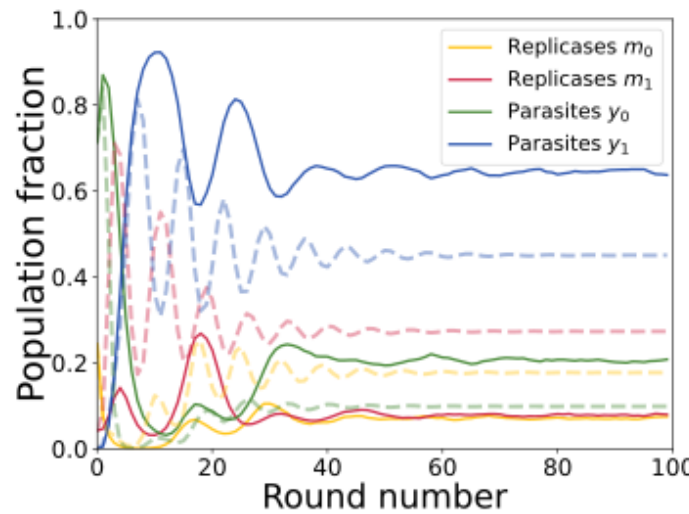
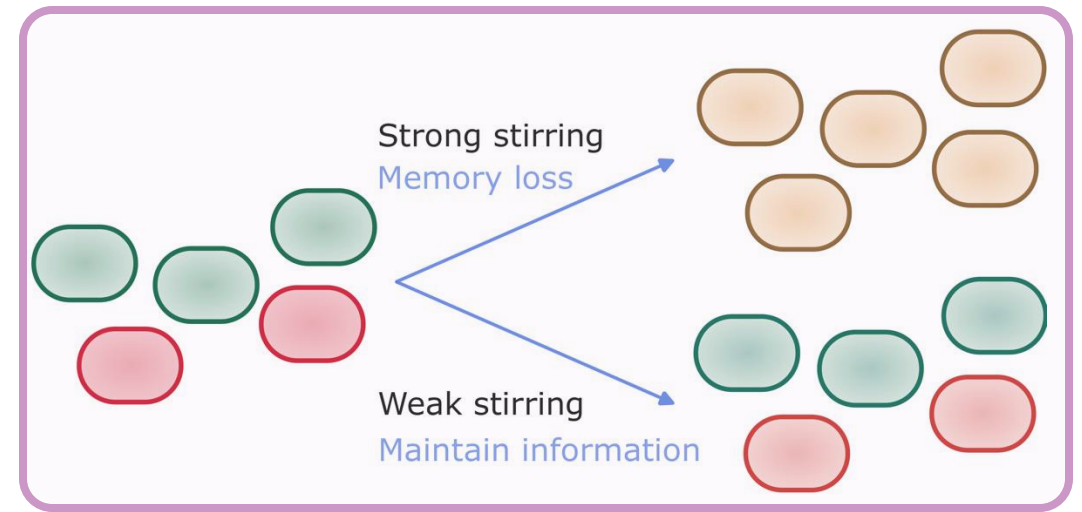


Round number

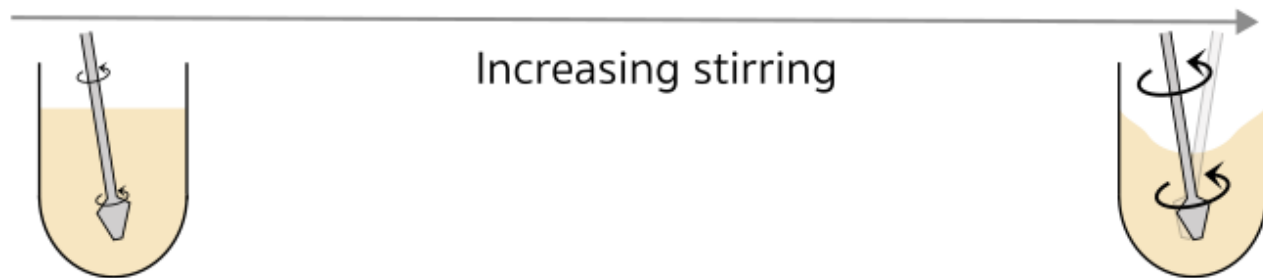
Stirring in serial transfer



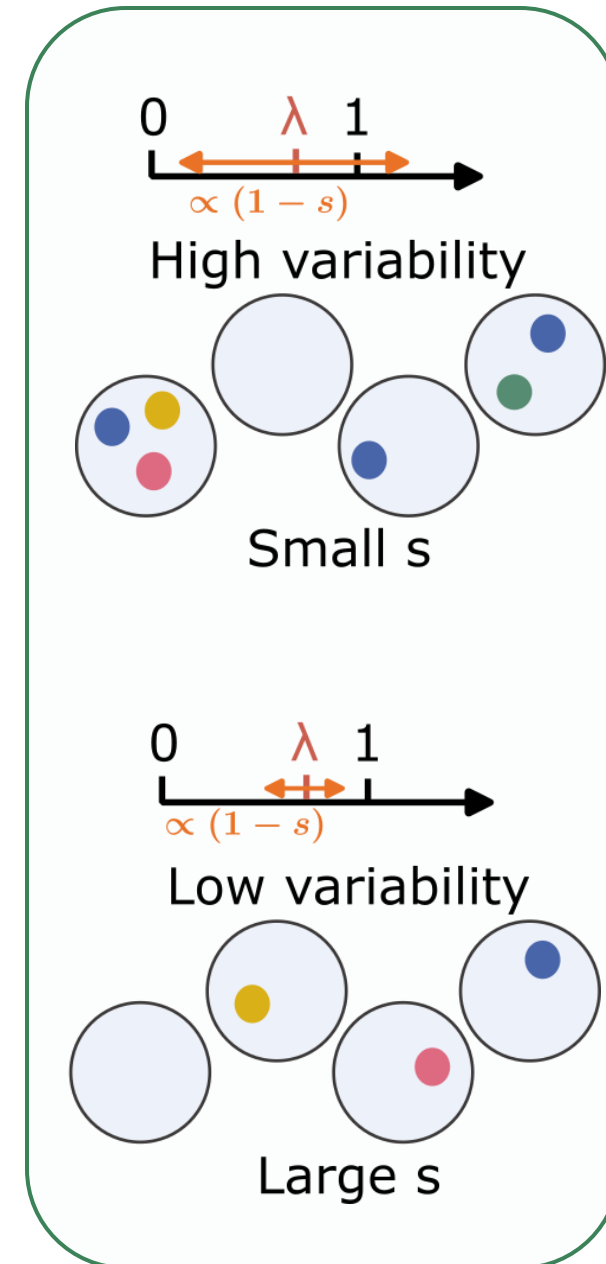
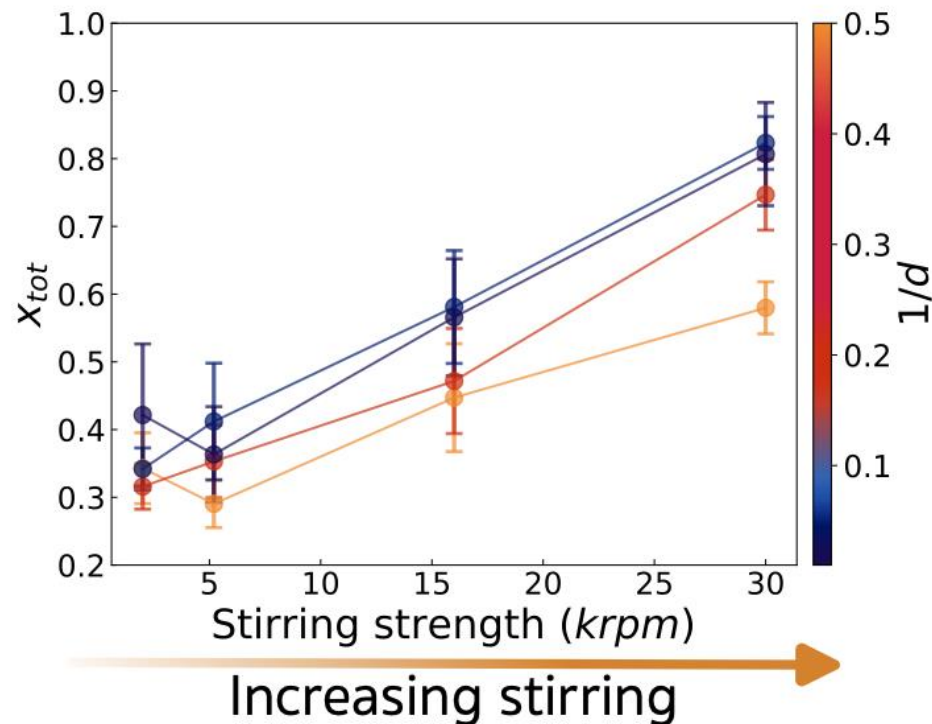
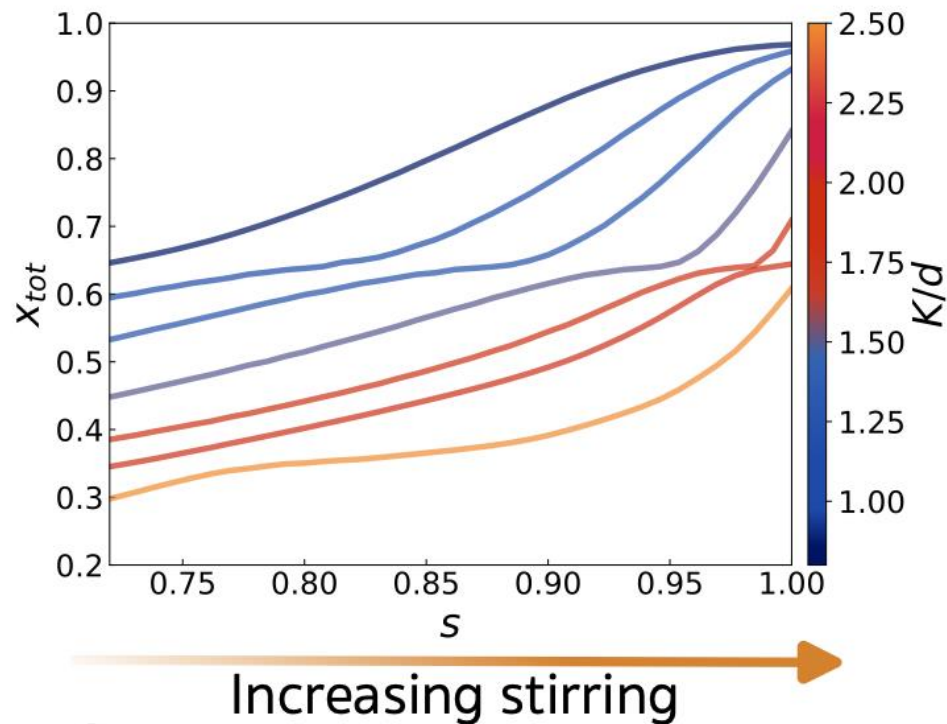
Tested experimentally



Increasing stirring



Stirring favors replicases



Conclusion

- **Compartments** along with **stirring** are a way to ensure **coevolution** and appearance of **complexity**.
- More species and strong dilution **stabilizes** the system.
- **Stirring** homogenizes the system, which tend to **isolate parasites** and favors **replicases**.



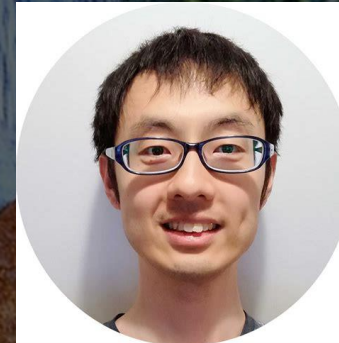
David Lacoste



Ryo Mizuuchi



Ryoka Kuwabara



Taro Furubayashi



Philippe Nghe

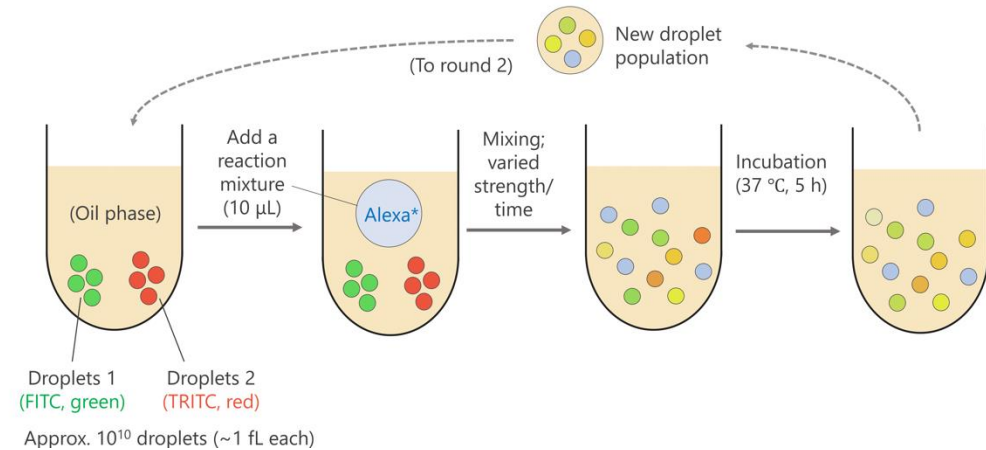
Additional content



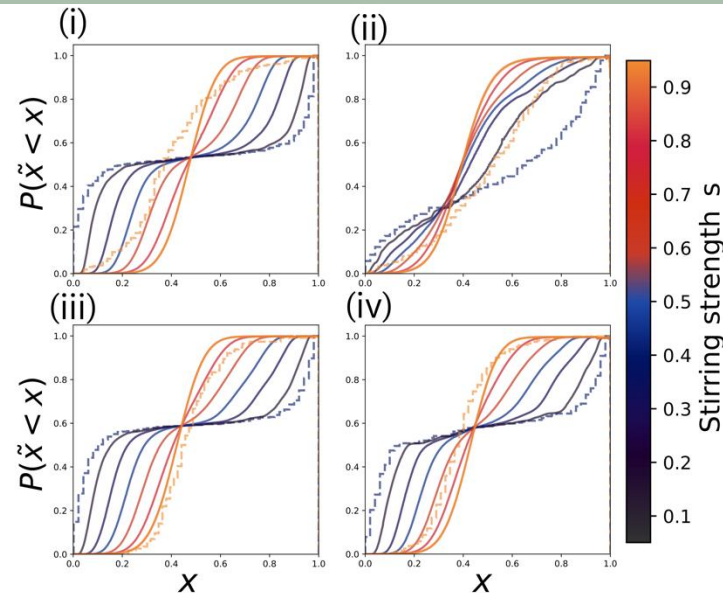
Experimental tests

With 2 species :

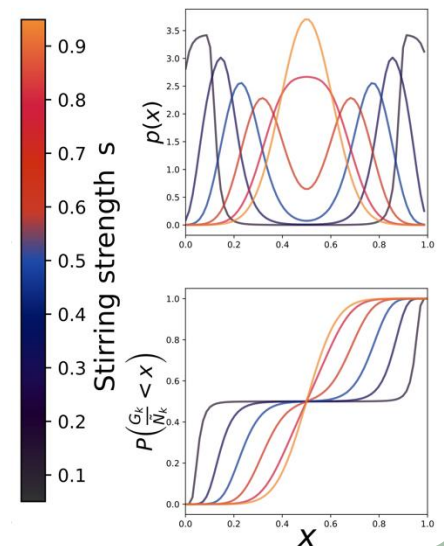
- From bimodal to monomodal distribution
- Depends on stirring process



- (i). Quick, strong stirring
- (ii). Long, weak stirring
- (iii). Medium length, strong stirring
- (iv). Long, strong stirring

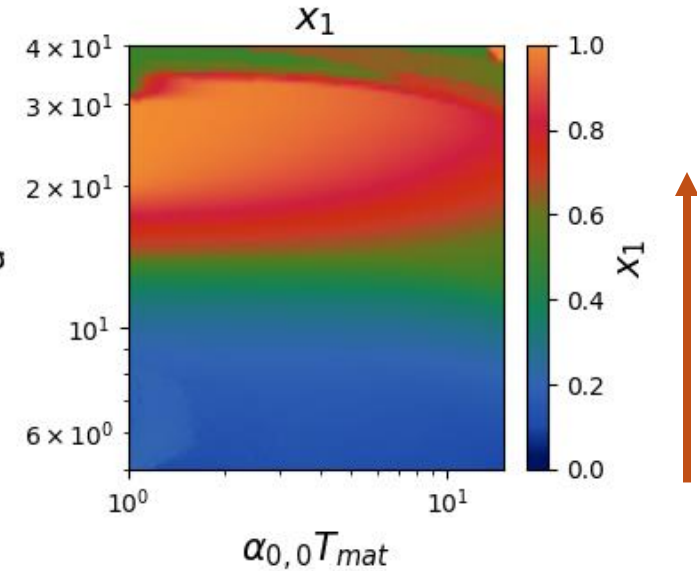
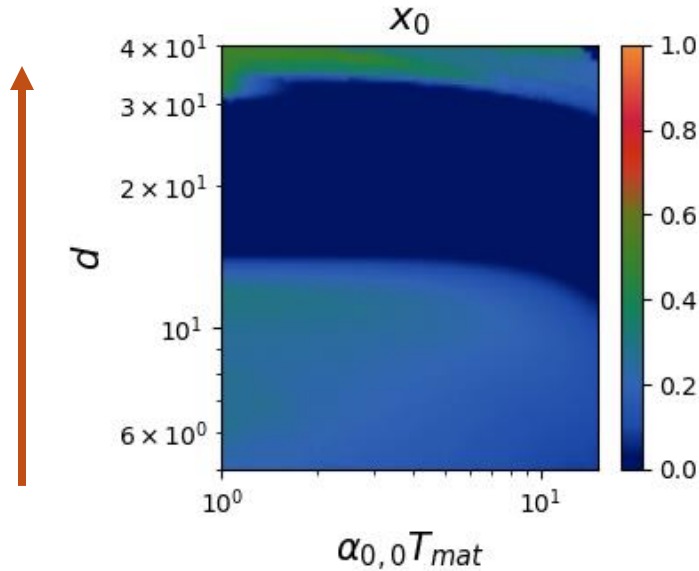


Distribution starting from 2 Dirac peaks.

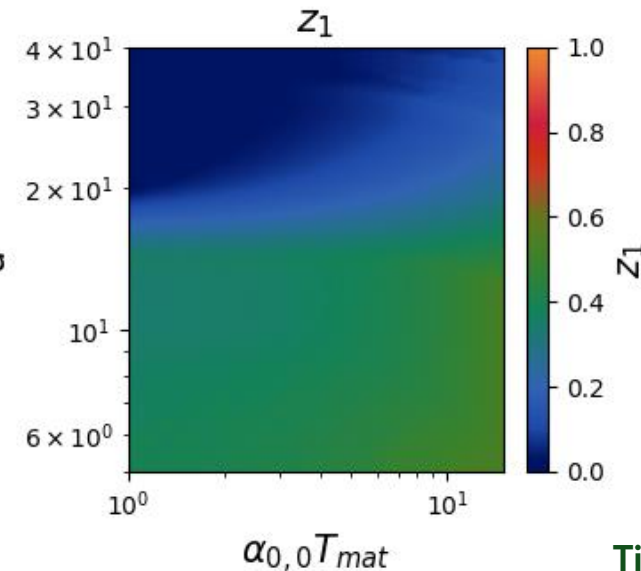
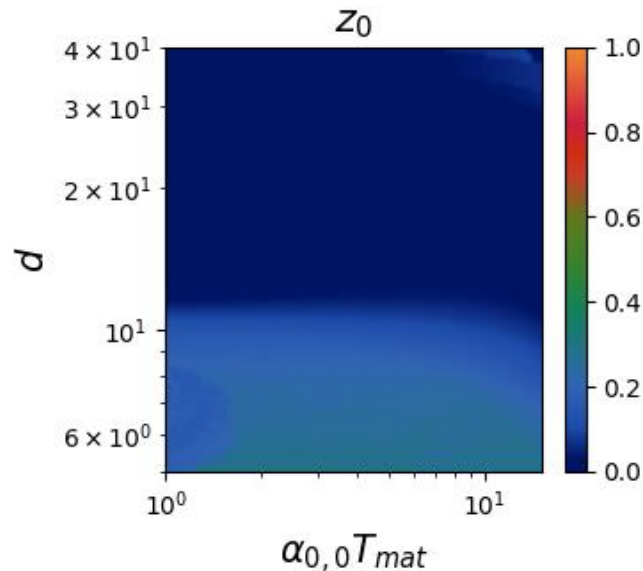


Influence of dilution factor and maturation time

Very strong dilution:
 $\lambda \leq 1 \Rightarrow$ No competition



Strong dilution:
 $\lambda \sim 1 \Rightarrow$ Isolate parasites



Time for maturation:
Mutations towards parasites